

DEBRECEN, Hungary

WMO: 128820

Lat: 47.489N

Lon: 21.615E

Elev: 361

StdP: 14.51

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.8	15.5	4.7	7.1	15.3	8.9	8.8	18.5	22.1	33.3	20.3	31.1	4.7	50	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	91.0	70.2	88.0	69.1	85.1	67.9	72.7	85.8	71.0	83.6	69.5	81.7	6.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.4	105.8	78.5	66.7	99.5	76.8	65.3	94.7	75.3	36.5	86.2	35.0	84.1	33.8	81.6	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
19.2	16.9	14.9	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			4.8	94.7	6.0	2.6	0.5	96.5	-3.1	98.0	-6.4	99.5	-10.8	101.4
			4.1	75.4	5.9	1.5	-0.1	76.5	-3.6	77.4	-6.9	78.3	-11.1	79.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.0	30.7	34.4	42.6	53.5	62.1	68.2	71.2	71.0	62.0	52.2	42.6	32.7	
	DBStd	16.18	8.26	8.71	7.93	7.11	6.38	6.06	5.47	5.65	6.43	7.47	8.03	8.36	
	HDD50	2181	599	436	248	46	2	0	0	0	1	66	246	537	
	HDD65	5360	1064	856	694	347	133	36	11	14	132	400	672	1001	
	CDD50	2922	0	0	19	151	377	546	658	652	362	132	24	1	
	CDD65	626	0	0	0	2	43	132	204	200	43	2	0	0	
	CDH74	6238	0	0	0	49	478	1266	1988	1993	436	28	0	0	
	CDH80	2135	0	0	0	6	95	406	741	776	111	0	0	0	
Wind	WSAvg	6.6	6.8	7.1	8.0	7.4	6.9	6.4	6.0	5.7	6.1	5.8	6.2	6.5	
Precipitation	PrecAvg	22.0	1.2	1.3	1.3	1.8	2.4	2.9	2.6	2.0	1.7	1.5	1.8	1.7	
	PrecMax	33.3	3.3	4.6	5.0	6.4	6.3	6.9	7.0	4.9	4.8	5.7	4.6	4.1	
	PrecMin	12.6	0.2	0.0	0.1	0.2	0.3	0.7	0.2	0.3	0.0	0.0	0.0	0.0	
	PrecStd	4.4	0.7	0.9	0.9	1.0	1.5	1.5	1.6	1.2	1.2	1.2	1.0	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.9	59.3	68.2	79.1	86.3	91.5	94.8	95.4	88.8	77.3	66.7	54.2	
		MCWB	46.8	50.3	53.4	61.2	65.4	71.7	70.3	69.7	66.5	62.8	57.3	49.6	
	2%	DB	48.0	53.6	63.7	74.4	82.7	88.0	90.5	91.6	82.9	73.6	62.0	49.8	
		MCWB	45.0	47.2	50.8	57.9	64.4	69.9	70.5	69.7	64.5	60.3	54.5	46.8	
	5%	DB	44.6	49.7	59.4	70.8	79.5	84.8	87.3	87.9	79.1	69.3	58.0	46.2	
		MCWB	42.1	44.7	48.9	56.2	63.4	68.1	69.4	69.0	63.4	58.9	52.2	43.9	
	10%	DB	41.7	46.2	55.4	67.3	75.9	81.5	84.2	84.3	75.1	64.8	54.3	43.5	
		MCWB	39.9	42.5	47.1	54.2	61.7	66.9	68.4	68.1	61.9	56.8	50.4	41.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.8	51.1	55.2	62.3	68.1	74.3	74.6	74.6	68.5	64.3	58.0	50.0
			MCDB	50.6	57.5	64.5	75.9	80.2	86.6	86.4	88.3	81.8	73.5	64.7	53.4
2%		WB	45.1	47.8	52.3	59.6	66.2	71.9	72.5	72.3	66.5	61.8	55.5	46.7	
		MCDB	47.7	52.9	60.7	71.2	78.5	84.1	85.6	85.4	78.9	71.1	60.6	49.4	
5%		WB	42.5	45.3	50.1	57.6	64.6	70.0	71.0	70.4	64.8	59.7	53.0	44.3	
		MCDB	44.3	48.8	57.4	67.9	76.6	81.4	83.9	83.3	76.0	67.5	57.1	45.9	
10%		WB	39.9	43.1	47.9	55.6	63.0	68.2	69.6	68.9	63.0	57.5	50.5	41.9	
		MCDB	41.5	45.8	54.8	64.9	73.8	79.0	81.5	81.5	72.4	64.1	54.1	43.4	
Mean Daily Temperature Range	5% DB	MDBR	9.9	13.1	17.0	19.7	19.9	19.6	20.4	21.2	19.6	18.8	13.8	9.4	
		MCDBR	12.7	18.1	23.5	25.3	25.3	24.2	25.3	26.4	25.8	24.6	19.5	12.6	
		MCWBR	10.1	12.4	13.6	12.3	10.9	9.9	9.4	9.5	10.5	13.2	12.6	9.9	
	5% WB	MCDBR	12.2	16.3	21.0	22.6	22.6	22.0	22.8	23.4	22.5	21.7	17.3	11.6	
		MCWBR	9.9	11.7	12.9	11.5	10.7	9.9	9.5	9.7	9.9	12.4	11.6	9.4	
Clear-Sky Solar Irradiance	taub	0.333	0.358	0.396	0.450	0.433	0.434	0.453	0.450	0.417	0.386	0.360	0.333		
	taud	2.322	2.282	2.213	2.085	2.201	2.226	2.181	2.187	2.244	2.310	2.329	2.346		
	Ebn at Noon	231	251	258	254	265	264	257	252	248	237	218	215		
	Edh at Noon	24	30	38	48	44	43	45	43	37	30	24	21		
All-Sky Solar Radiation	RadAvg	317	588	1002	1452	1773	1912	1871	1705	1189	766	411	243		
	RadStd	40	96	134	138	168	120	160	150	134	84	40	37		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	-223	N/A	+54
(47)	Regional (52 neighbors)	+1.19	N/A	N/A	+1.80	+0.85	N/A	-225	-342	+198	+99

Nomenclature: See separate page